Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $b > a$, and $c > 0$ then the equation $(x-a)(x-b) - c = 0$ has:

(A) both roots in $(-\infty, a)$

(B) both roots in $(a, b]$

(C) one root in $(-\infty, a)$ and other root in (b, ∞)

(D) one root in $(-\infty, a)$ and other root in $[a, b]$

Q.2

If the quadratic equation $x^2 + 2(k+1)x + 9k - 5 = 0$ has exactly one positive root, then k lies in the set

(A) $[5/9, \infty)$

(B) $(-\infty, 1) \cup (6, \infty)$

(C) $(-\infty, 5/9]$

(D) $[1, 6]$

Q.3

If $k \in \mathbf{R}$ lies between the roots of $ax^2 + 2bx + c = 0$, then

(A) $ak^2 + 2bk + c < 0$

(B) $a^2k^2 + 2abk + ac < 0$

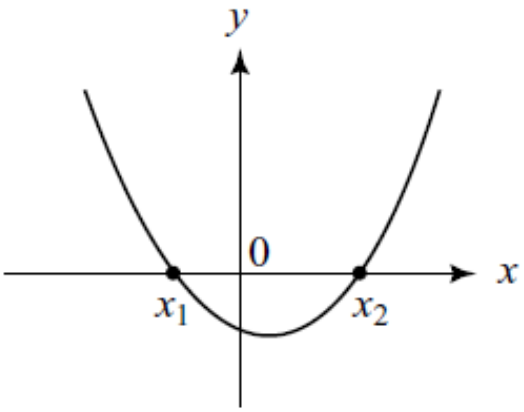
(C) $a^2k^2 + 2abk + ac > 0$

(D) $ak^2 + 2bk + c > 0$

Q.4

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Fig. Shows graph of $y = ax^2 + bx + c$. Then which one of the following is not true.



(A) $a > 0$

(B) $c < 0$

(C) $b^2 - 4ac > 0$

(D) $b > 0$

Q.5

If $a > 0$ and both the roots of $ax^2 + bx + c = 0$ are more than 1, then

(A) $a + b + c > 0$

(B) $a + b + 4c = 0$

(C) $a + b + c < 0$

(D) $a + 4b + c = 0$

Q.6

If α and β are the roots of $4x^2 - 16x + t = 0, \quad \forall t > 0$ such that $1 < \alpha < 2 < \beta < 3$, then the number of integral values of t are

(A) 2

(B) 3

(C) 4

(D) 5

Q.7

If $p, q, r, s \in$ R, then the equation $x^2 + px + 3q\text{-}x^2 + rx + q\text{-}x^2 + sx - 2q = 0$ has

(A) 6 real roots

(B) at least two real roots

(C) 2 real and 4 imaginary roots

(D) 4 real and 2 imaginary roots

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Q.8

For the equation x^2 - $2x$ - $3 = b$, which of the following statements is true?

(A) For $b < 0$, there are no solutions

(B) For $b = 0$, there are three solutions

(C) For $0 < b < 4$, there are two solutions

(D) For $b = 4$, there are four solutions

Q.9

Let $\mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{d}$ be distinct real numbers such that a, b are roots of $\mathrm{x}^2 - 5\mathrm{cx} - 6\ \mathrm{d} = 0$, and c, d are roots of $x^2 - 5ax - 6b = 0$. Then $b + d$ is

(A) 180

(B) 162

(C) 144

(D) 126

Q.10

If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are real and less than 3, then

(A) $a < 2$

(B) $2 \leq a \leq 3$

(C) $3 < a \leq 4$

(D) $a > 4$

Q.11

If the equation $x^2 + 2(k + 1)x + 9k - 5 = 0$ has only negative roots, then

(A) $k \leq 0$

(B) $k \geq 0$

(C) $k \geq 6$

(D) $k \leq 6$

Q.12

If $f\left(x_1\right) - f\left(x_2\right) = f\left(\frac{x_1 - x_2}{1 - x_1 x_2}\right)$ for $x_1, x_2 \in (-1, 1)$
then what is $\mathrm{f}(\mathrm{x})$ equal to?

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(A) $\ln\left(\frac{1-x}{1+x}\right)$

(B) $\ln\left(\frac{2+x}{1-x}\right)$

(C) $\tan^{-1}\left(\frac{1-x}{1+x}\right)$

(D) $\tan^{-1}\left(\frac{1+x}{1-x}\right)$

Q.13

If $\mathrm{A}=\left\{\mathrm{x} \in \mathrm{IR}: \mathrm{x}^2+6\mathrm{x}-7<0\right\}$ and
$\mathrm{B}=\left\{\mathrm{x} \in \mathrm{IR}: \mathrm{x}^2+9\mathrm{x}+14>0\right\}$, then which of the following is/ are correct?
1 $(\mathrm{A} \cap \mathrm{B})=(-2,1)$
2 $(\mathrm{A} \cup \mathrm{B})=(-7,-2)$
Select the correct answer using the code given below:

(A) 1 only

(B) 2 Only

(C) Both 1 and 2

(D) Neither 1 nor 2

Q.14

Let $a>0, b>0$ and $c>0$. Then both the roots of the equation $ax^2+bx+c=0$

(A) are real and negative

(B) have negative real parts

(C) are rational numbers

(D) None of these

Q.15

The roots of the equation $\left|\ x^2-x-6\ \right|=x+2$ are

(A) $(-2,1,4)$

(B) $(0,2,4)$

(C) $(0,1,4)$

(D) $(-2,2,4)$

Answers & Solutions

Q.1 Answer:

one root in $(-\infty, a)$ and other root in (b, ∞)

Solution:

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Let $f(x) = (x-a)(x-b) - c$
Use coefficient of $x^2 > 0$ $f(a) = -c = f(b) < 0$.

Q.2 Answer:

$(-\infty, 5/9]$

Solution:

One $+ve$ root means other root is 0 or negative this means product of roots is non-positive.
$D \geq 0$ and product of roots $\leq 0 \Rightarrow (k+1)^2 - (9k-5) \geq 0$ and $9k - 5 \leq 0$
$\Rightarrow k^2 - 7k + 6 \geq 0$ and $\leq 5/9$
$\Rightarrow (k-1)(k-6) \geq 0$ and $k \leq 5/9$
Thus, $k \leq 5/9$

Q.3 Answer:

$a^2k^2 + 2abk + ac < 0$

Solution:

Use : $(k-\alpha)(k-\beta)$ have the opposite signs, where α, β are roots of $ax^2 + 2bx + c = 0$.
$\left(k^2 - k\beta - k\alpha + \alpha\beta\right) < 0$
$\left(k^2 + \frac{k2b}{a} + \frac{c}{a}\right) < 0$
$a^2k^2 + 2abk + ac < 0$

Q.4 Answer:

$b > 0$

Solution:

As parabola open upwards, $a > 0$. Also, for $x = 0 ax^2 + bx + c = c < 0$.
c < 0
the product of $x1 \times 2 < 0$ as they have opposite signs.
Since $\times 1 \times 2 = c/a, c/a < 0$.
Since $c < 0$ a must be > 0 for the condition c/a<0 to hold.
Answer should be (1) $a > 0$.

Q.5 Answer:

$a + b + c > 0$

Solution:

Suppose $\alpha, \beta > 1, \alpha < \beta$ be roots of $ax^2 + bx + c = 0$, then $ax^2 + bx + c > 0$ if $x < \alpha$ or $x > \beta < 0$ if $\alpha < x < \beta$
Thus, $a + b + c > 0$.

Q.6 Answer:

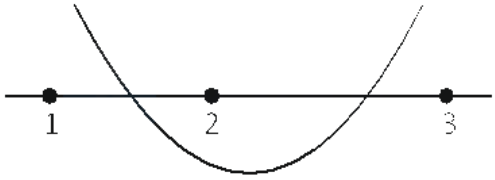
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Solution:

$fx = 4x^2 - 16x + t$

Opening upward parabola



$f1 > 0, \quad f2 < 0, \quad f3 > 0$

$f1 > 0 \Rightarrow 4 - 16 + t > 0 \Rightarrow t > 12$

$f2 < 0 \Rightarrow 16 - 32 + t < 0 \Rightarrow t < 16$

$f3 > 0 \Rightarrow 36 - 48 + t > 0 \Rightarrow t > 12$

$\Rightarrow 12 < t < 16 \Rightarrow t = 13, \quad 14, \quad 15$

Hence, the number of integral values $t = 13,14,15$ **is** 3.

Q.7 Answer:

at least two real roots

Solution:

For real roots,

$p^2 - 12q \geq 0$ **...(i)**

$r^2 + 4q \geq 0$ **...(ii)**

$s^2 - 8q \geq 0$ **...(iii)**

If $q > 0$**, then (ii) is always true.**

If $q < 0$**, then (i) and (iii) are always true.**

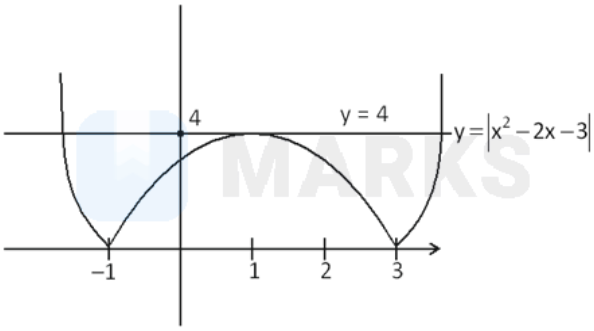
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So, the equation has atleast two real roots.

Q.8 Answer:

For $b < 0$, there are no solutions

Solution:



$b < 0 \rightarrow$ no solution

$b = 0 \rightarrow$ two solutions

$0 < b < 4 \rightarrow$ four solutions

$b = 4 \rightarrow$ three solutions

Q.9 Answer:

144

Solution:

Here, a and b are roots of the equation $x^2 - 5cx - 6d = 0$

$\therefore \mathrm{a} + \mathrm{b} = \frac{-(-5)\mathrm{c}}{1} = 5\mathrm{c}$ **...(1)**

And

$\Rightarrow \mathrm{a} \cdot \mathrm{b} = \frac{-6\,\mathrm{d}}{1} = -6\,\mathrm{d} \quad \ldots\ (2)$

Now, c and d are roots of the equation $x^2 - 5ax - 6d = 0$

$\therefore \mathrm{c} + \mathrm{d} = \frac{-(-5\mathrm{a})}{1} = 5\mathrm{a} \ldots (3)$

$\Rightarrow \mathrm{cd} = \frac{-6\,\mathrm{b}}{1} = -6\,\mathrm{b} \ldots (4)$

Adding (1) and (2),

$\Rightarrow \mathrm{a} + \mathrm{b} + \mathrm{c} + \mathrm{d} = 5\mathrm{c} + 5\mathrm{a}$

$\Rightarrow \mathrm{b} + \mathrm{d} = 4\mathrm{c} + 4\mathrm{a}$

$\Rightarrow \mathrm{b} + \mathrm{d} = 4(\mathrm{c} + \mathrm{a})$

Now dividing (1) by (2) we get, $\Rightarrow \frac{a+b}{c+d} = \frac{5c}{5a}$

$\Rightarrow 5\mathrm{a}^2 + 5\mathrm{ab} = 5\mathrm{c}^2 + 5\ \mathrm{cd}$

$\Rightarrow 5\mathrm{a}^2 + 5(-6\ \mathrm{d}) = 5\mathrm{c}^2 + 5(-6\ \mathrm{b})$

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$\Rightarrow 5a^2 - 30\,d = 5c^2 - 30\,b$

$\Rightarrow 5\left(a^2 - c^2\right) = -30\,b + 30\,d$

$\Rightarrow (a + c)(a - c) = -6\,b + 6\,d$

$\Rightarrow (a + c)(a - c) = -6(5c - a) + 6(5a - c)$

$\Rightarrow (a + c)(a - c) = -30c + 6a + 30a - 6c$

$\Rightarrow (a + c)(a - c) = -36c + 36a$

$\Rightarrow (a + c)(a - c) = 36(a - c)$

$\Rightarrow a + c = 36$ **Substituting** $\Rightarrow b + d = 4(36)$

$\therefore b + d = 144$

Substituting value of $a + c$ **in equation (5) we get,**

$\Rightarrow b + d = 4(36)$

$\therefore b + d = 144$

Q.10 Answer:

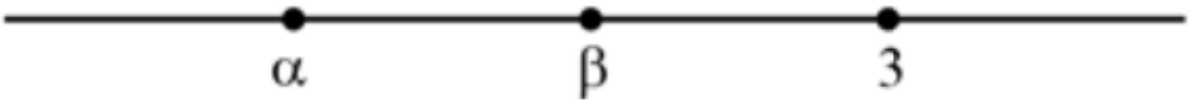
$a < 2$

Solution:

$f(x) = x^2 - 2ax + a^2 + a - 3 = 0$

$f(3) > 0, \alpha + \beta < 6, \Delta \geq 0$

$\Rightarrow a^2 - 5a + 6 > 0, a < 3, -4a + 12 > 0 \Rightarrow a < 2$ **or** $a > 3 \Rightarrow a < 2$



Q.11 Answer:

$k \geq 6$

Solution:

Let $f(x) = x^2 + 2(k + 1)x + 9k - 5$

Let α, β **be the roots of** $f(x) = 0$ **The equation** $f(x) = 0$ **will have both negative roots, if**

(i) Disc ≥ 0

(ii) $\alpha < 0, \beta < 0$ **i.e.,** $\alpha + \beta < 0$ **and (iii)** $\alpha.\beta > 0$ **Now, Disc** $\geq 0 \Rightarrow \quad 4(k + 1)^2 - 36k + 20 > 0 \Rightarrow k^2 - 7k + 6 \geq 0 \Rightarrow (k - 1)(k - 6) \geq 0$

$\Rightarrow \quad k \leq 1$ **or** $k \geq 6$

. . . (i) $\alpha + \beta < 0 \Rightarrow -2(k + 1) < 0 \Rightarrow k + 1 > 0 \Rightarrow k > -1$

and $\alpha.\beta > 0 \Rightarrow 9k - 5 > 0 \Rightarrow k > \frac{5}{9}$

from (i), (ii) and (iii), we get $k \geq 6$

Q.12 Answer:

$\ln\left(\frac{1-x}{1+x}\right)$

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Solution:

$f(x_1) - f(x_2) = f\left(\frac{x_1 - x_2}{1 - x_1 x_2}\right)$

$x_1, x_2 \in (-1, 1)$

then $f(x) = \log \frac{(1-x)}{(1+x)}$

$f(x_1) = \log \frac{1-x_1}{1+x_1}$

$f(x_2) = \log \dfrac{1 - x_2}{1 + x_2}$

$$
\begin{aligned}
f(x_1) - f(x_2) &= \log \frac{1 - x_1}{1 + x_1} - \log \frac{1 - x_2}{1 + x_2} \\
&= \log \frac{(1 - x_1)}{(1 + x_1)} \times \frac{(1 + x_2)}{(1 - x_2)} \\
&= \log \frac{(1 - x_1 + x_2 - x_1 x_2)}{(1 + x_1 - x_2 - x_1 x_2)} \\
&= \log \frac{(1 - x_1 x_2) - (x_1 - x_2)}{(1 - x_1 x_2) + (x_1 - x_2)} \\
&= \log \frac{1 - \left(\frac{x_1 - x_2}{1 - x_1 x_2}\right)}{1 + \left(\frac{x_1 - x_2}{1 - x_1 x_2}\right)}
\end{aligned}
$$

$f(x_1) - f(x_2) = f\left(\frac{x_1 - x_2}{1 - x_1 x_2}\right)$

Q.13 Answer:

Both 1 and 2

Solution:

$x^2 + 6x - 7 < 0$

$\Rightarrow (x + 7)(x - 1) < 0$

$\Rightarrow x = (-7, 1)$

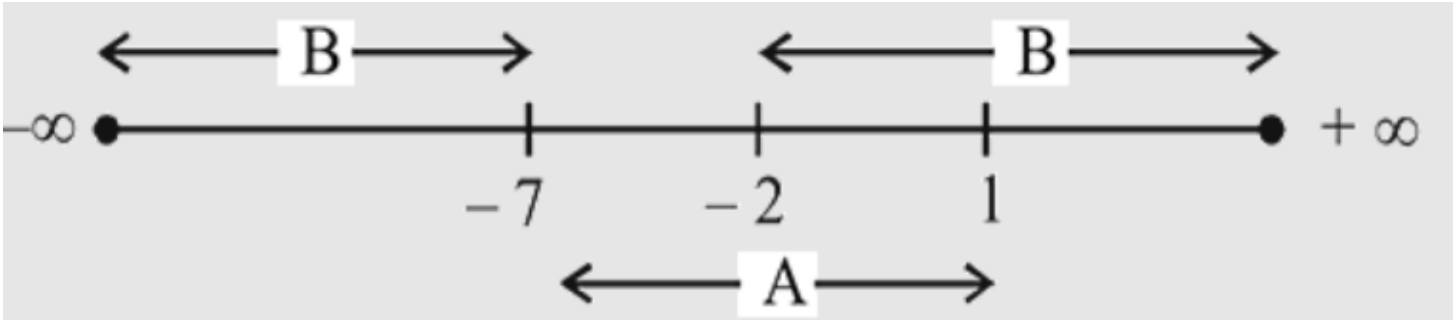
$\Rightarrow A = \{-6, -5, -4, -3, -2, -1, 0\}$

$\Rightarrow x^2 + 9x + 14 > 0$

$\Rightarrow (x + 7)(x + 2) > 0$

$\Rightarrow x = (-\infty, -7) \cup (-2, \infty)$

$\Rightarrow B = R - \{-7, -6, -5, -4, -3, -2\}$



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So $A \cap B = (-2, 1)$
$A \cup B = (-7, -2)$

Q.14 Answer:

have negative real parts

Solution:

Let $a > 0, b > 0, c > 0$
Given equation $ax^2 + bx + c = 0$
we know that $\mathrm{D} = b^2 - 4ac$ and
$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Let $b^2 - 4ac > 0, b > 0$
If $a > 0, c > 0$ then $b^2 - 4ac < b^2$
$\Rightarrow$ Roots are negative
Let $b^2 - 4ac < 0$, then $x = \frac{-b \pm i\sqrt{4ac - b^2}}{2a}$
$\Rightarrow$ roots are imaginary and have negative real part.
$(\because b > 0)$

Q.15 Answer:

$(-2, 2, 4)$

Solution:

We have, $\left|x^2 - x - 6\right| = x + 2$ Now, we have to consider two cases, Case I: When $x \leq -2$ or $x \geq 3 \therefore \left|x^2 - x - 6\right| = x + 2$

$\Rightarrow \quad x^2 - x - 6 = x + 2 \Rightarrow x^2 - 2x - 8 = 0$ or $x^2 - 4x + 2x - 8 = 0$

$\Rightarrow \quad (x - 4)(x + 2) = 0 \Rightarrow x = -2, 4$

Case II: When $-2 < x < 3$

$\Rightarrow \quad -\left(x^2 - x - 6\right) = x + 2 \Rightarrow -x^2 + x + 6 = x + 2 \Rightarrow -x^2 = -4$

$\Rightarrow \quad x^2 = 4 \Rightarrow x = \pm 2$

Hence, the roots are $(-2, 2, 4)$

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